

ALvl Chem 18 EQ P1 22w to 09s Paper 1 Carboxylic acids & derivatives 94marks

As you start and work through this worksheet you can tick off your progress to show yourself how much you have done, and what you need to do next. The first task is just to read the first question and should take you less than one minutes to complete.

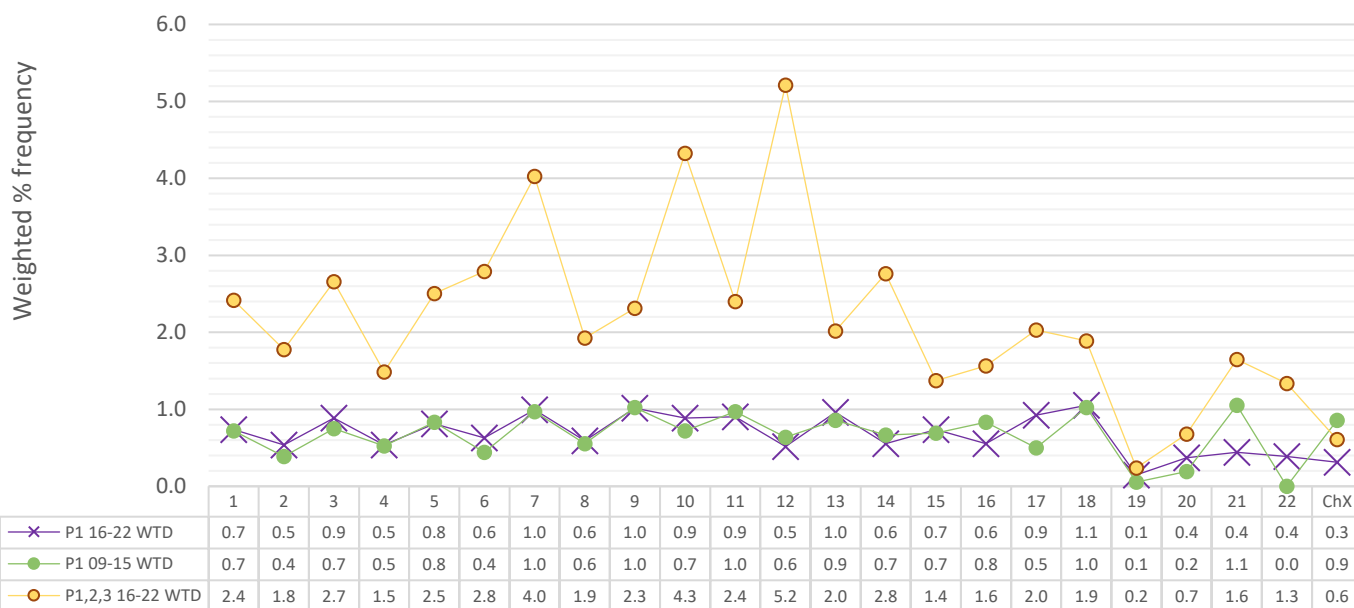
Paper 1 Topic 18

Checklist Tick each task off as you go along

RANK:

P1 Noob	P1 Novice	P1 Bronze	P1 Silver	P1 Gold	P1 ¹ Winner	P1 Hero	P1 Legend
1 Q started	1 Q done	10% of marks	25% of marks	40% of marks	50% of marks	75% of marks	100% of marks
	1	9	24	38	47	71	94
	1	12	29	47	59	88	118

9701 Chemistry Weighted Mark Frequency: Paper 1 Topics
2022w to 2016m in purple crosses, compared to earlier and all AS exams combined



What the most thoughtful students will get out of their extensive studying will be a capacity to do meaningful brain-based work even under stressful conditions, which is a part of the self-mastery skillset that will continue to deliver value for the whole of their lives. Outstanding grades will also happen, but the most important goal from skillful action in study is being better at any important task, even if circumstances do not feel ideal.

As you are moving through your studies you can learn more about yourself by trying out new ways to manage yourself, and analysing how effective those new techniques were. In this reflective process not only will you get better at working positively and productively to deliver ambitious and successful outcomes, but you will be working towards one aspect of life's highest pursuit, summarised and inscribed on the Temple of Apollo at Delphi: "know thyself".

1. To complete these questions, as important as your answer, is checking your answer against the mark scheme.
2. For each page or group of 10 questions, convert your mark score into a percentage. This will allow you to see (and feel) your progress as you get more experience and understanding with each topic.
3. Multiple choice questions, done carefully where you explain and show yourself your thinking using written notes as you move through each question, can be more useful than just Paper 2 for students aiming for a C or B grade. Paper 2 should be the larger focus for students aiming for A and A* grades, however.
4. If you find you get a higher percentage answering short answer questions than multiple choice questions that often means you are NOT using the marking scheme correctly; your correct answer might not be fully complete for all the marks you are awarding. The marks easiest to miss rely on providing the largest amount of detail.

¹ **DO NOT** work on these higher levels of completion in your A2 year unless you have also achieved at least a "Silver" (25%) in the same topic in **Paper 2**, which is **MOST** of your **AS grade**, and Paper 3 which is a smaller part of your year but still important.

18 Carboxylic acids and derivatives

18.1 Carboxylic acids

Learning outcomes

Candidates should be able to:

- recall the reactions by which carboxylic acids can be produced:
 - oxidation of primary alcohols and aldehydes with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and refluxing
 - hydrolysis of nitriles with dilute acid or dilute alkali followed by acidification
 - hydrolysis of esters with dilute acid or dilute alkali and heat followed by acidification
- describe:
 - the redox reaction with reactive metals to produce a salt and $\text{H}_2(\text{g})$
 - the neutralisation reaction with alkalis to produce a salt and $\text{H}_2\text{O}(\text{l})$
 - the acid-base reaction with carbonates to produce a salt and $\text{H}_2\text{O}(\text{l})$ and $\text{CO}_2(\text{g})$
 - esterification with alcohols with concentrated H_2SO_4 as catalyst
 - reduction by LiAlH_4 to form a primary alcohol

18.2 Esters

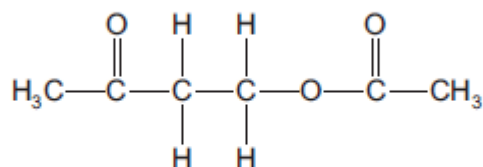
Learning outcomes

Candidates should be able to:

- recall the reaction (reagents and conditions) by which esters can be produced:
 - the condensation reaction between an alcohol and a carboxylic acid with concentrated H_2SO_4 as catalyst
- describe the hydrolysis of esters by dilute acid and by dilute alkali and heat

Q# 1128/ AS Chemistry/2022/w/TZ 1/Paper 1/Q# 36//www.SmashingScience.org :o)

- 36 Compound X reacts with ethanoic acid in the presence of an H^+ catalyst to produce the compound shown.



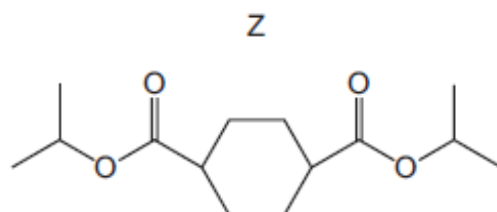
What is the molecular formula of compound X?

- A $\text{C}_2\text{H}_4\text{O}$ B $\text{C}_2\text{H}_6\text{O}_2$ C $\text{C}_4\text{H}_8\text{O}$ D $\text{C}_4\text{H}_8\text{O}_2$

Q# 1129/ AS Chemistry/2022/w/TZ 1/Paper 1/Q# 32//www.SmashingScience.org :o)

- 32 An alcohol, X, reacts with a dicarboxylic acid, Y, to form a double ester, Z.

The diagram shows the structure of the ester.



Which row about the reactants forming ester Z is correct?

	the class of alcohol X	the shape of the ring in Y
A	secondary	non-planar
B	secondary	planar
C	tertiary	non-planar
D	tertiary	planar

Q# 1130/ AS Chemistry/2022/s/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

38 Two 1g samples of Y are reacted separately and completely with sodium and with sodium carbonate. The volumes of the gases produced are collected and measured.

	relative volumes of gases	
	with Na	with Na ₂ CO ₃
Y	2	1

What could Y be?

- A $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{OH}$
- B $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$
- C $\text{CH}_3\text{COCH}_2\text{OH}$
- D $\text{CH}_3\text{COCO}_2\text{H}$

Q# 1131/ AS Chemistry/2022/s/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

37 Which compound produces butan-2-ol and ethanoic acid on hydrolysis?

- A $\text{CH}_3\text{CO}_2\text{CH}(\text{CH}_3)_2$
- B $\text{CH}_3\text{CO}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
- C $\text{CH}_3\text{CH}(\text{CH}_3)\text{CO}_2\text{CH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$

Q# 1132/ AS Chemistry/2022/s/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 How many esters have the molecular formula $\text{C}_4\text{H}_8\text{O}_2$?

- A 2 B 3 C 4 D 5

Q# 1133/ AS Chemistry/2022/s/TZ 1/Paper 1/Q# 27//www.SmashingScience.org :o)

27 Which compound will react with LiAlH_4 to form two optical isomers?

- A $\text{CH}_3\text{CH}_2\text{COCH}_3$
- B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- C $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CO}_2\text{H}$



35 Compound X contains a single ester group.

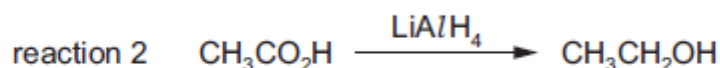
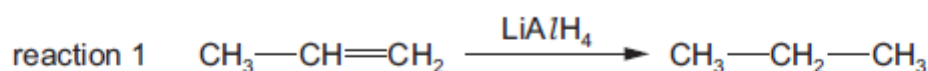
X contains 27.6% by mass of oxygen.

Which pair of products could be produced by the hydrolysis of X?

- A butan-1-ol and ethanoic acid
- B ethanol and propanoic acid
- C methanol and methanoic acid
- D propan-2-ol and butanoic acid

Q# 1135/ AS Chemistry/2022/m/TZ 2/Paper 1/Q# 34//www.SmashingScience.org :o)

34 A student suggests two uses of LiAlH_4 .

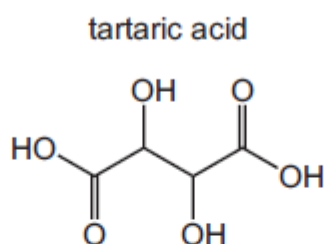


Which reactions would give the product shown?

- A both reactions
- B reaction 1 only
- C reaction 2 only
- D neither reaction

Q# 1136/ AS Chemistry/2022/m/TZ 2/Paper 1/Q# 32//www.SmashingScience.org :o)

32 The structure of tartaric acid is shown.



Four moles of substance X react with one mole of tartaric acid.

What could be substance X?

- A sodium
- B sodium carbonate
- C sodium hydrogencarbonate
- D sodium hydroxide



29 The structural formula of an ester is $(\text{CH}_3)_2\text{CHOCO}(\text{CH}_2)_2\text{CH}_3$.

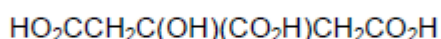
This ester is boiled with aqueous hydrochloric acid.

Which two products are formed?

- A propan-1-ol and butanoic acid
- B propan-2-ol and butanoic acid
- C propan-1-ol and propanoic acid
- D propan-2-ol and propanoic acid

28 Citric acid is found in lemon juice.

citric acid



Which volume of 0.40 mol dm^{-3} sodium hydroxide solution is required to neutralise a solution containing 0.0050 mol of citric acid?

- A 12.5 cm^3 B 25.0 cm^3 C 37.5 cm^3 D 50.0 cm^3

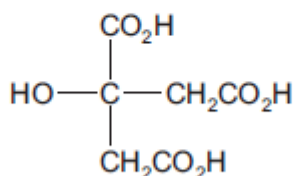
24 When an organic compound, Q, is treated with phosphorus pentachloride, fumes of hydrogen chloride are evolved. When Q is warmed with acidified aqueous potassium dichromate(VI), the solution turns green.

What is Q?

- A $\text{CH}_3\text{CH}_2\text{CHO}$
- B $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
- C $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- D $(\text{CH}_3)_3\text{COH}$

27 How many moles of hydrogen, H_2 , are evolved when an excess of sodium metal is added to one mole of citric acid?

citric acid



- A 0.5 B 1.5 C 2 D 4



24 The compound cetyl palmitate, $C_{15}H_{31}CO_2C_{16}H_{33}$, is a waxy solid.

Cetyl palmitate is heated under reflux with an excess of aqueous sodium hydroxide.

Which products will be formed?

- A $C_{15}H_{31}ONa$ and $C_{16}H_{33}CO_2Na$
- B $C_{15}H_{31}CO_2Na$ and $C_{16}H_{33}ONa$
- C $C_{15}H_{31}OH$ and $C_{16}H_{33}CO_2Na$
- D $C_{15}H_{31}CO_2Na$ and $C_{16}H_{33}OH$

Q# 1142/ AS Chemistry/2021/m/TZ 2/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 An organic compound, T, does **not** fizz when aqueous sodium carbonate is added to it.

Compound T contains 27.6% by mass of oxygen.

What could be the identity of T?

- 1 propanal
- 2 ethyl butanoate
- 3 3-methylpentanoic acid

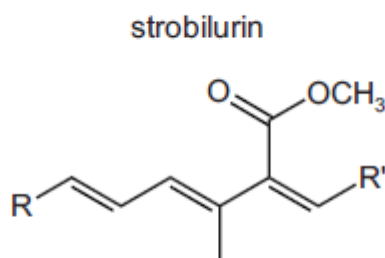
Q# 1143/ AS Chemistry/2021/m/TZ 2/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Which reaction gives butanoic acid as one of its products?

- A acid hydrolysis of butyl ethanoate
- B alkaline hydrolysis of butyl ethanoate
- C acid hydrolysis of ethyl butanoate
- D alkaline hydrolysis of ethyl butanoate

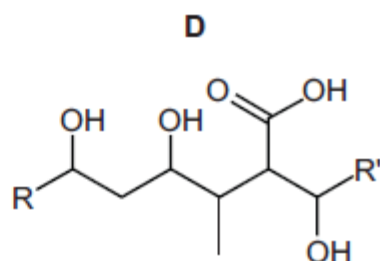
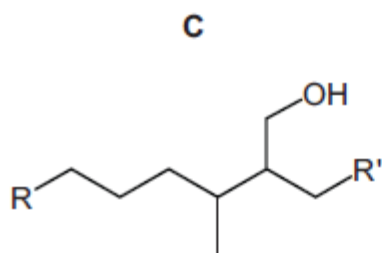
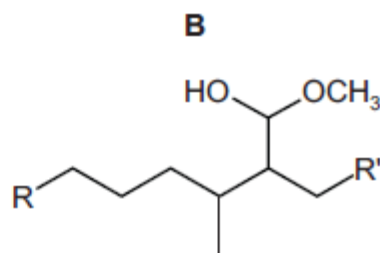
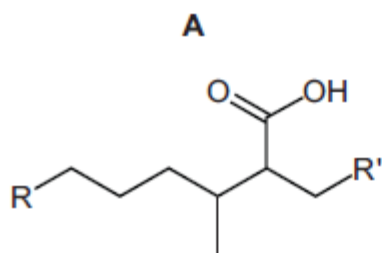
Q# 1144/ AS Chemistry/2021/m/TZ 2/Paper 1/Q# 23//www.SmashingScience.org :o)

23 Part of the structure of strobilurin is shown. R and R' are inert groups.



Strobilurin is warmed with aqueous sulfuric acid producing compound X. Compound X is then treated with hydrogen in the presence of a nickel catalyst producing compound Y.

What could be the structure of compound Y?

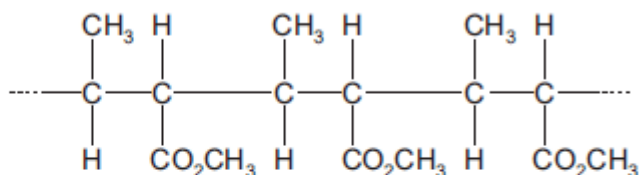


Q# 1145/ AS Chemistry/2020/w/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 The diagram shows part of the structure of polymer X.

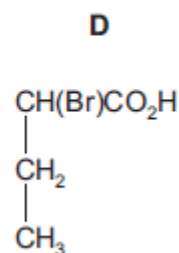
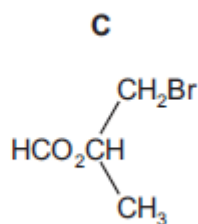
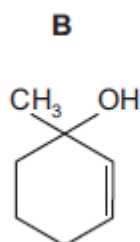
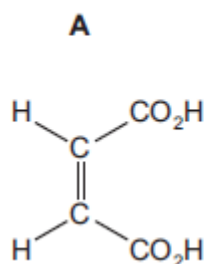


Which reagents react with polymer X?

- 1** aqueous sulfuric acid
- 2** aqueous sodium hydroxide
- 3** sodium

Q# 1146/ AS Chemistry/2020/w/TZ 1/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Which compound is chiral and reacts with Na_2CO_3 to give CO_2 ?

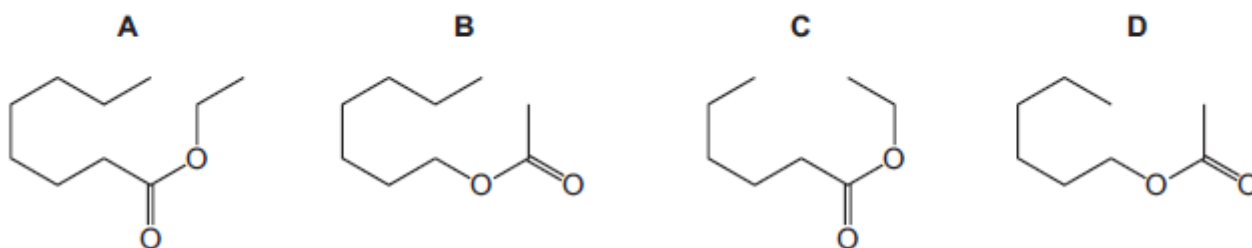


30 Which reaction produces an organic anion with a good yield?

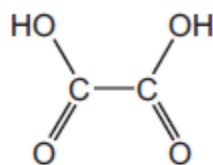
- A heating ethanenitrile under reflux with dilute sodium hydroxide
- B heating ethanenitrile under reflux with dilute sulfuric acid
- C heating ethane with sodium metal
- D heating ethanol under reflux with dilute sodium hydroxide

26 When compound X is heated under reflux with aqueous sodium hydroxide solution two products are formed: sodium ethanoate and hexan-1-ol.

What is compound X?



22 The diagram shows the structure of ethanedioic acid.



Ethanedioic acid reacts with ethanol in the presence of a few drops of concentrated sulfuric acid to form a diester. The molecular formula of the diester is $C_6H_{10}O_4$.

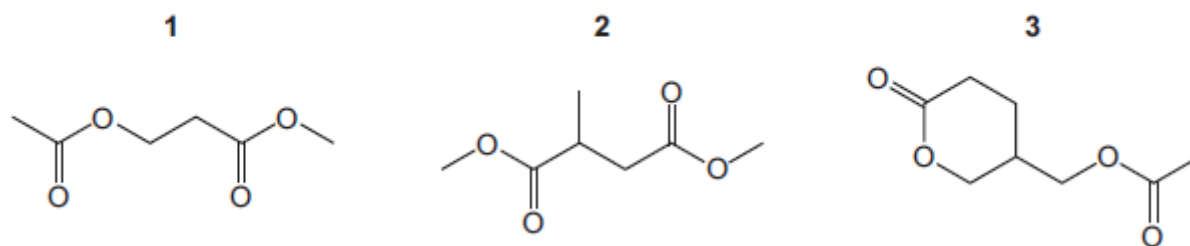
What is the structural formula of the diester?

- A $CH_3CH_2CO_2CO_2CH_2CH_3$
- B $CH_3CH_2OCOCO_2CH_2CH_3$
- C $CH_3CH_2O_2CO_2CCH_2CH_3$
- D $CH_3CO_2CH_2CH_2OCOCH_3$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

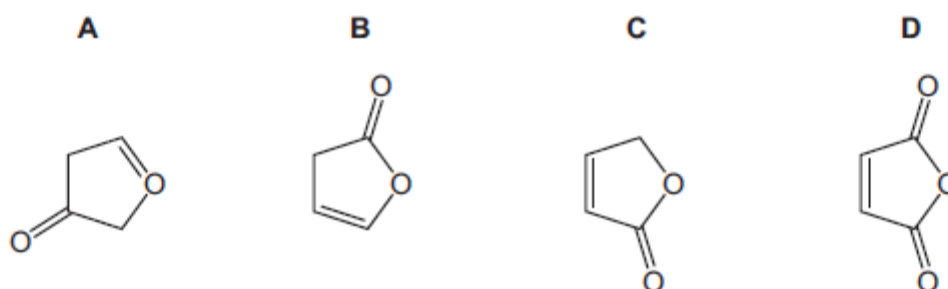
40 Which compounds produce three **different** organic products when hydrolysed?



Q# 1151/ AS Chemistry/2019/w/TZ 1/Paper 1/Q# 29//www.SmashingScience.org :o)

29 When $\text{CH}_2(\text{OH})\text{CH}=\text{CHCO}_2\text{H}$ is warmed with a little concentrated sulfuric acid, a cyclic compound is formed.

What is the skeletal formula of the cyclic compound?



Q# 1152/ AS Chemistry/2019/w/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 Tartaric acid, $\text{HO}_2\text{CCH}(\text{OH})\text{CH}(\text{OH})\text{CO}_2\text{H}$, is found in many plants.

A sample of tartaric acid reacts with an excess of LiAlH_4 to form the organic product J.

What happens when $\text{NaOH}(\text{aq})$ is added to separate samples of tartaric acid and J?

- A Both tartaric acid and J react.
- B Only tartaric acid reacts.
- C Only J reacts.
- D Neither tartaric acid nor J react.

Q# 1153/ AS Chemistry/2019/s/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 Carboxylic acids react with alcohols to produce esters.

Carboxylic acid X forms **one** ester only with molecular formula $C_5H_{10}O_2$.

What could X be?

- 1 ethanoic acid
- 2 propanoic acid
- 3 butanoic acid

Q# 1154/ AS Chemistry/2019/s/TZ 1/Paper 1/Q# 39//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

39 Carboxylic acids can be made by several different reactions.

Which statements are correct?

- 1 The acid hydrolysis of CH_3CH_2CN will make ethanoic acid.
- 2 The oxidation of $CH_3CH_2CH_2CH_2OH$ will make butanoic acid.
- 3 The oxidation of CH_3CH_2CHO will make propanoic acid.

Q# 1155/ AS Chemistry/2019/m/TZ 2/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 Carboxylic acids can be prepared from alcohols, nitriles or esters.

Which statements are correct?

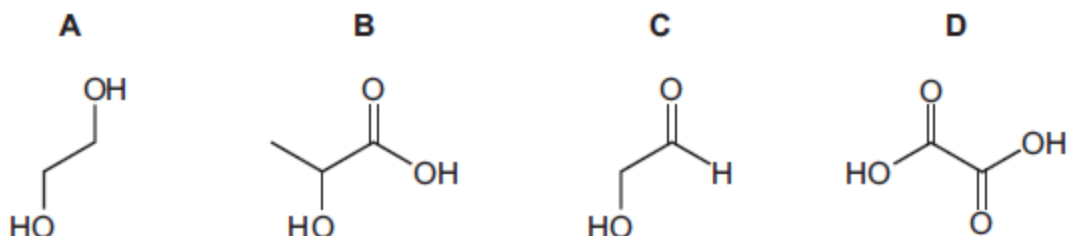
- 1 Both primary and secondary alcohols can be oxidised to carboxylic acids.
- 2 Carboxylic acids can be made from nitriles by hydrolysis.
- 3 Ethyl propanoate gives propanoic acid when reacted with hydrochloric acid.



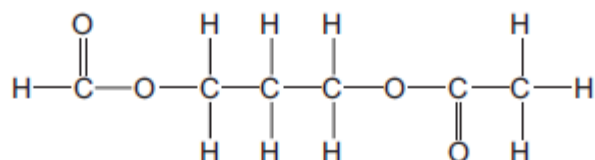
29 1 mole of each of the following four compounds is reacted separately with:

- an excess of sodium
- an excess of sodium carbonate.

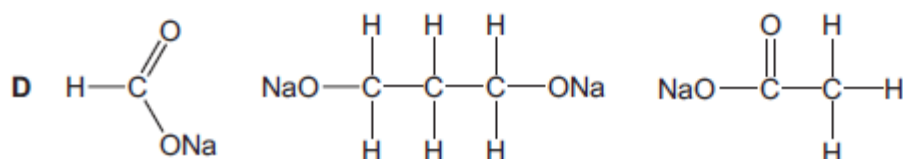
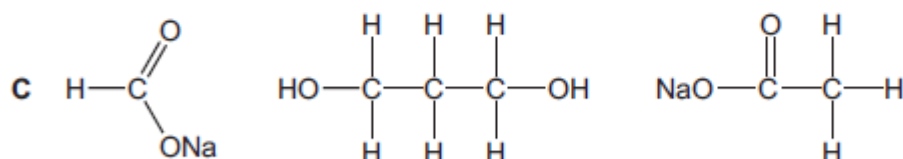
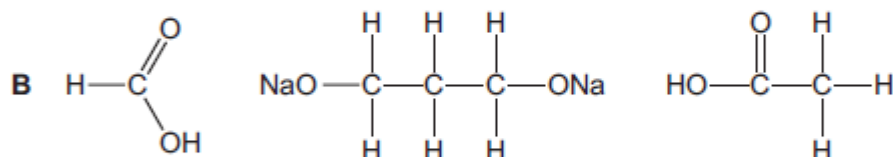
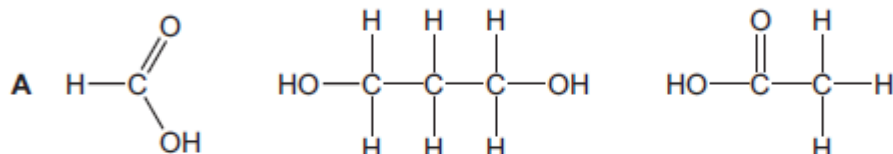
Which compound produces the same volume of gas with each of the **two** reagents?



29 The diester shown can be hydrolysed by heating with an excess of aqueous sodium hydroxide.



What would the products of this reaction be?



28 Ethanedioic acid, $\text{HO}_2\text{CCO}_2\text{H}$, is reduced using an excess of lithium aluminium hydride, LiAlH_4 .

What is the organic product of the reaction?

- A ethanol
- B ethane-1,2-diol
- C ethanedial, OHCCHO
- D methane

Q# 1159/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 26//www.SmashingScience.org :o)

26 Sodium reacts with 1 mol of compound Y to produce 1 mol of $\text{H}_2(\text{g})$.

Which compound could Y be?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- B $(\text{CH}_3)_3\text{COH}$
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
- D $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$

Q# 1160/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 An organic molecule W contains 3 carbon atoms. It requires 4.5 molecules of oxygen for complete combustion.

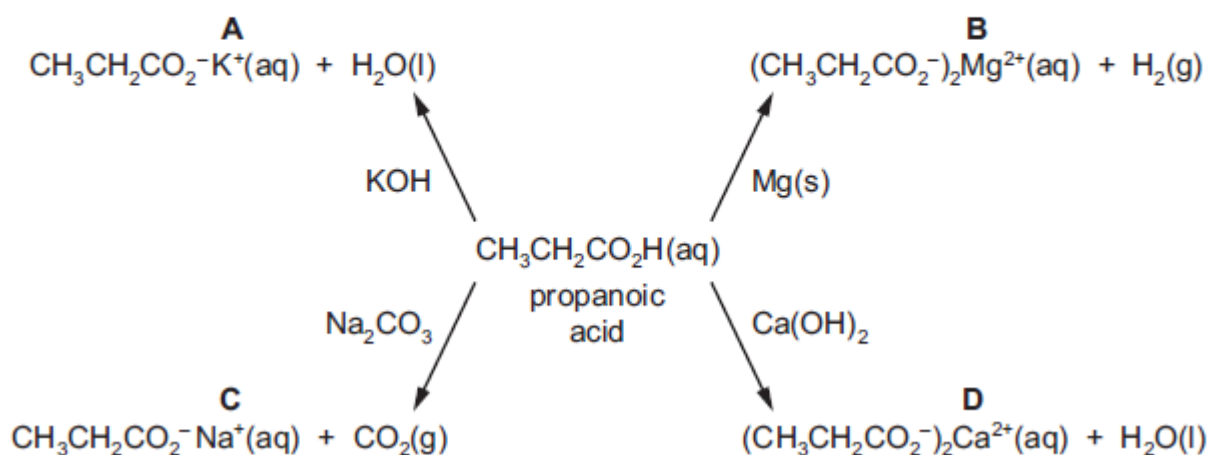
What could W be?

- A propane
- B propanoic acid
- C propanone
- D propan-1-ol

Q# 1161/ AS Chemistry/2018/s/TZ 1/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Four reactions of propanoic acid to form salts and other products are shown.

Which reaction does **not** show the formulae of **all** the correct products?



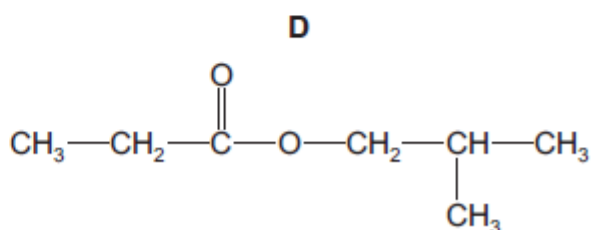
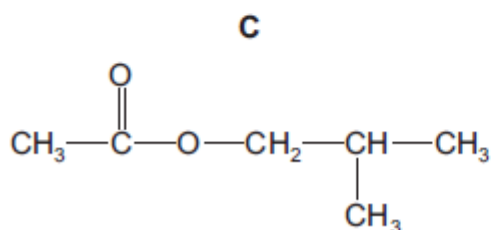
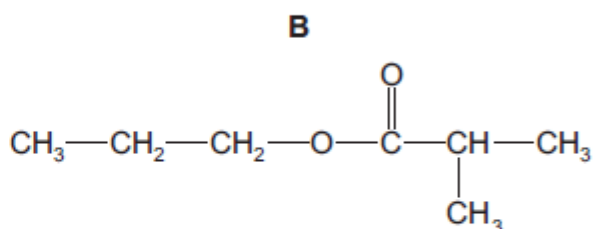
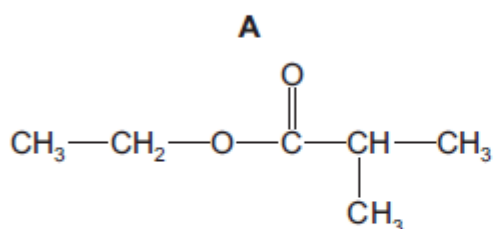
28 Which reagent could be used to carry out the following reaction?



- A a solution containing acidified dichromate(VI) ions
- B a solution containing dilute, acidified manganate(VII) ions
- C a solution containing hot, concentrated, acidified manganate(VII) ions
- D concentrated sulfuric acid

27 Ethyl propanoate is refluxed with aqueous sodium hydroxide. The alcohol produced is then reacted with methyl propanoic acid to make a second ester.

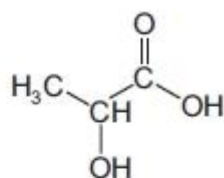
What is the structural formula of this second ester?



The responses **A** to **D** should be selected on the basis of

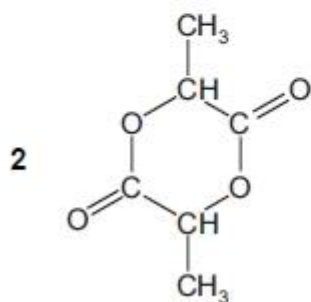
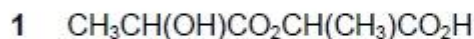
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 The structure of lactic acid is shown.



lactic acid

Which esters might form when lactic acid is heated?



Q# 1165/ AS Chemistry/2018/m/TZ 2/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Alcohols, aldehydes and nitriles can each be converted into carboxylic acids.

Which descriptions of their conversions into carboxylic acids are correct?

	alcohols	aldehydes	nitriles
A	hydrolysis	hydrolysis	hydrolysis
B	hydrolysis	hydrolysis	oxidation
C	oxidation	oxidation	hydrolysis
D	oxidation	oxidation	oxidation

Q# 1166/ AS Chemistry/2018/m/TZ 2/Paper 1/Q# 28//www.SmashingScience.org :o)

28 Ethanedioic acid has the formula $\text{HO}_2\text{CCO}_2\text{H}$.

What is the formula of aluminium ethanedioate?



Q# 1167/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 30//

30 A sample of the ester $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ is hydrolysed. The product mixture is then treated with hot, acidified KMnO_4 .

What are the final carbon-containing products?

- A $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ only
B $\text{CH}_3\text{CO}_2\text{H} + \text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
C $\text{CH}_3\text{CO}_2\text{H} + \text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
D $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$

28 At room temperature, propanoic acid was reacted to produce sodium propanoate. No gas was produced during the reaction.

What could the propanoic acid have reacted with?

- A** $\text{NaHCO}_3(\text{aq})$ **B** $\text{NaOH}(\text{aq})$ **C** $\text{Na}_2\text{CO}_3(\text{aq})$ **D** $\text{Na}_2\text{SO}_4(\text{aq})$

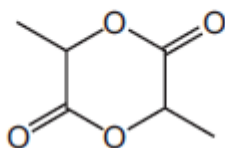
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 Which statements about ethanol and ethanoic acid are correct?

- 1** Both react with a suitable reagent to form an ester.
- 2** Both react with sodium.
- 3** Both are soluble in water.

24 Lactide is an intermediate in the manufacture of a synthetic fibre.



lactide

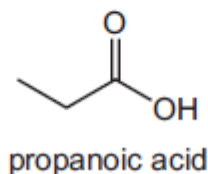
Which compound, on heating with an acid catalyst, can produce lactide?

- A** hydroxyethanoic acid
B 2-hydroxybutanoic acid
C 2-hydroxypropanoic acid
D 3-hydroxypropanoic acid

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 40 Propanoic acid occurs naturally as a result of the bacterial fermentation of milk and is partly responsible for the flavour of Swiss cheese.

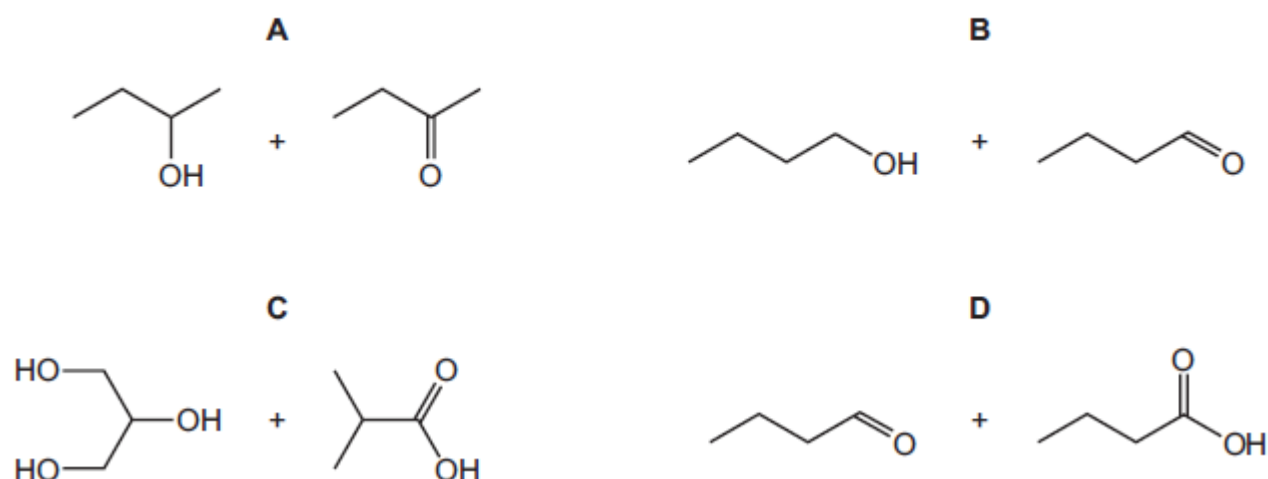


Which starting materials can be used to produce propanoic acid?

- 1 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- 2 $\text{CH}_3\text{CH}_2\text{CHO}$
- 3 $\text{CH}_3\text{CH}_2\text{CN}$

Q# 1172/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 30//www.SmashingScience.org :o)

- 30 Which two compounds can react together to produce an ester?

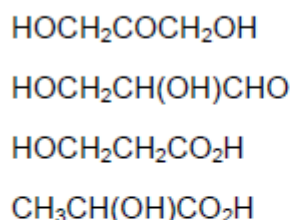


Q# 1173/ AS Chemistry/2016/w/TZ 1/Paper 1/Q# 39//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 39 Several structural isomers of $\text{C}_3\text{H}_6\text{O}_3$ are listed below.

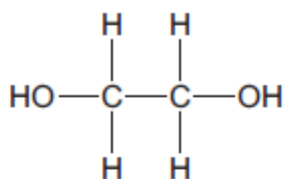


Which statements about these structural isomers are correct?

- 1 One mole of each reacts with two moles of sodium.
- 2 Only one of the isomers contains a tertiary alcohol group.
- 3 They all contain a primary alcohol group.

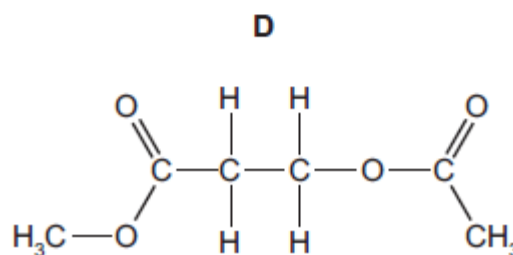
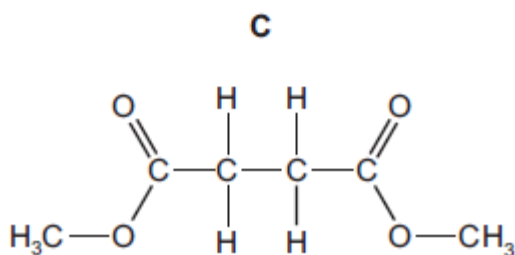
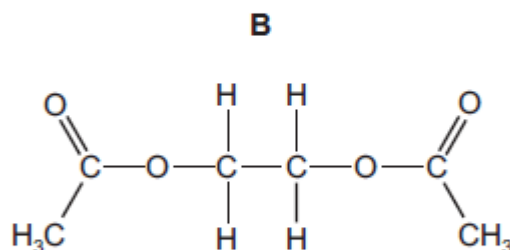
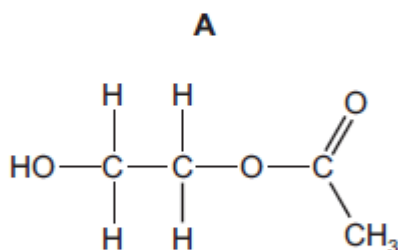


- 30 A solvent, X, used in printing inks has a molecular formula $C_6H_{10}O_4$. It may be made by reacting ethane-1,2-diol with ethanoic acid in the presence of an acid catalyst.

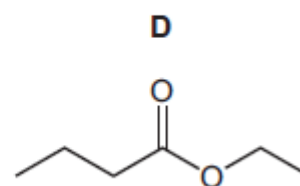
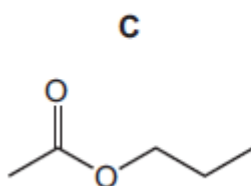
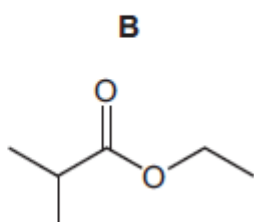
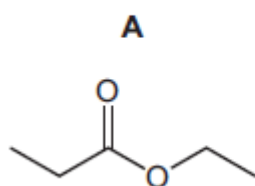


ethane-1,2-diol

What is the structure of solvent X?



- 29 Which formula represents an ester that will form propanoic acid on hydrolysis with dilute sulfuric acid?

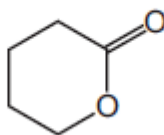


- 29 The ester $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$ was hydrolysed under acidic conditions.

What are the organic products of this hydrolysis?

- A butanoic acid and 2-methylpropan-1-ol
- B butanoic acid and 2-methylpropan-2-ol
- C butan-1-ol and 2-methylpropanoic acid
- D propanoic acid and 2-methylpropan-1-ol

27 Cyclic esters are also known as lactones. *Delta* lactone is used as a solvent and in the manufacture of polyesters.

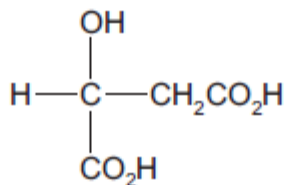


delta lactone

From which compound could *delta* lactone be made by a single reaction?

- A $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
- B $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
- C $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- D $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$

26 Malic acid is found in apples.

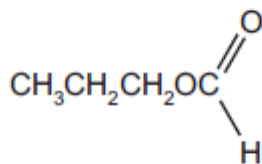


malic acid

Which reagent will react with all three -OH groups present in the malic acid molecule?

- A ethanol in the presence of concentrated sulfuric acid
- B potassium hydroxide
- C sodium
- D sodium carbonate

21 The structural formula of compound X is shown below.



compound X

What is the name of compound **X** and how does its boiling point compare with that of butanoic acid?

	name of X	boiling point of X
A	methyl propanoate	higher than butanoic acid
B	methyl propanoate	lower than butanoic acid
C	propyl methanoate	higher than butanoic acid
D	propyl methanoate	lower than butanoic acid

Q# 1180/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 39//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 39** An unknown organic compound **Z** reacts with sodium to give a combustible gas as one product. **Z** does **not** give a yellow precipitate with alkaline aqueous iodine.

What is a possible identity of **Z**?

- 1 ethanoic acid
- 2 pentan-3-ol
- 3 propan-1-ol

Q# 1181/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 37** Which reagents, when used in an excess, can be used to make sodium lactate, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{Na}$, from lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$?

- 1 Na
- 2 NaHCO_3
- 3 NaOH

Q# 1182/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 30//www.SmashingScience.org :o)

- 30** Which row of the table is correct?

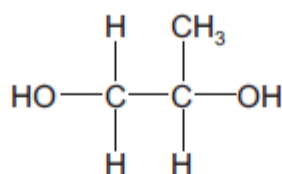
	increasing number of carbon atoms $\longrightarrow$			
A	ethyl methanoate	methyl propanoate	pentyl pentanoate	propyl butanoate
B	ethyl methanoate	methyl propanoate	propyl butanoate	pentyl pentanoate
C	methyl propanoate	propyl butanoate	ethyl methanoate	pentyl pentanoate
D	propyl butanoate	ethyl methanoate	pentyl pentanoate	methyl propanoate

28 Which mixture could be used to produce propyl methanoate?

- A $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ and CH_3OH
- B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ and HCO_2H
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and HCO_2H
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$ and CH_3OH

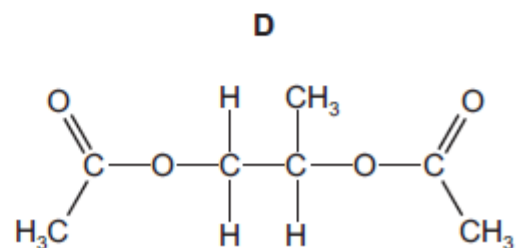
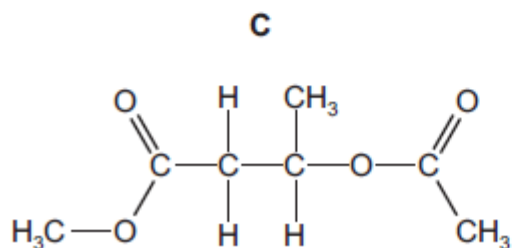
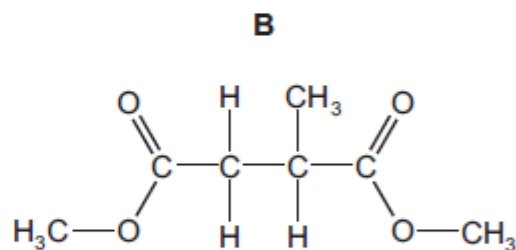
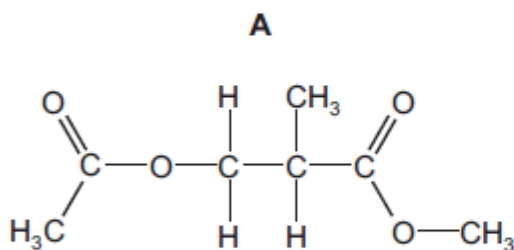
Q# 1184/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 27//www.SmashingScience.org :o)

27 The fragrance compounds of perfumes are often dissolved in solvent Y, which has a molecular formula $\text{C}_7\text{H}_{12}\text{O}_4$. It is made by reacting propane-1,2-diol with ethanoic acid in the presence of an acid catalyst.



propane-1,2-diol

What is the structure of solvent Y?



Q# 1185/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 39//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



39 Which reactions result in the formation of propanoic acid?

- 1 $\text{CH}_3\text{CH}_2\text{CO}_2\text{Na}$ with dilute $\text{H}_2\text{SO}_4(\text{aq})$
- 2 $\text{CH}_3\text{CH}=\text{CHCH}_3$ with hot, concentrated $\text{H}^+ / \text{MnO}_4^-(\text{aq})$
- 3 $\text{CH}_3\text{CH}_2\text{OH}$ with $\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}(\text{aq})$

Q# 1186/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

37 An oxidising agent that can oxidise ethanal to ethanoic acid, or to ethanoate ions, will also oxidise methanoic acid, HCO_2H , to carbon dioxide and water.

Which reagents, on heating, will react differently with HCO_2H and $\text{CH}_3\text{CO}_2\text{H}$?

- 1 $\text{Na}_2\text{CO}_3(\text{aq})$
- 2 Fehling's reagent
- 3 dilute acidified KMnO_4

Q# 1187/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 30//www.SmashingScience.org :o)

30 Which compound produces butan-2-ol and ethanoic acid on hydrolysis?

- A $\text{CH}_3\text{CO}_2\text{CH}(\text{CH}_3)_2$
- B $\text{CH}_3\text{CO}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
- C $\text{CH}_3\text{CH}(\text{CH}_3)\text{CO}_2\text{CH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$

Q# 1188/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 20//www.SmashingScience.org :o)

20 How many isomeric esters have the molecular formula $\text{C}_4\text{H}_8\text{O}_2$?

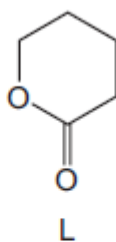
- A 2 B 3 C 4 D 5

Q# 1189/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 5-hydroxypentanoic acid is readily converted into the cyclic compound L.



Which statements about this reaction are correct?

- 1 Acidified sodium dichromate(VI) is used as a reagent.
- 2 A water molecule is produced in the reaction.
- 3 The reaction is catalysed by concentrated H_2SO_4 .

Q# 1190/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

38 An organic compound, X, will react with calcium metal to produce a salt with the empirical formula $\text{CaC}_4\text{H}_4\text{O}_4$.

What could be the identity of X?

- 1 ethanoic acid
- 2 butanedioic acid
- 3 2-methylpropanedioic acid

Q# 1191/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

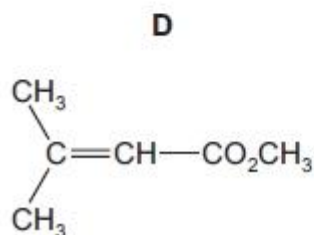
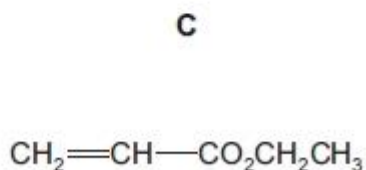
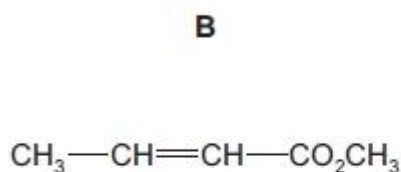
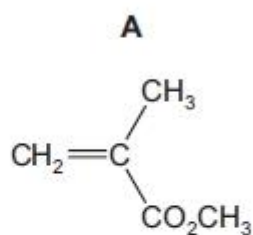
28 The ester $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_3$ is responsible for the aroma of apples.

When this ester is hydrolysed by acid in the stomach, what is the empirical formula of the organic acid produced?

- A CH_2O B CH_4O C $\text{C}_2\text{H}_4\text{O}$ D $\text{C}_3\text{H}_6\text{O}_2$

27 Methyl methylpropenoate is the monomer used to make *Perspex*.

Which diagram correctly shows methyl methylpropenoate?



Q# 1193/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 Use of the Data Booklet is relevant to this question.

In an organic synthesis, a 62% yield of product is achieved.

Which conversions are consistent with this information?

- 1 74.00g of butan-2-ol → 44.64g of butanone
- 2 74.00g of butan-1-ol → 54.56g of butanoic acid
- 3 74.00g of 2-methylpropan-1-ol → 54.56g of 2-methylpropanoic acid

Q# 1194/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



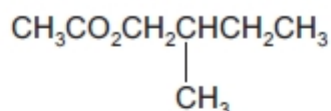
38 The molecule responsible for the pineapple flavour used in sweets is $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$.

Which statements about this molecule are correct?

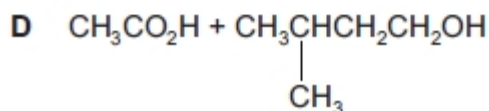
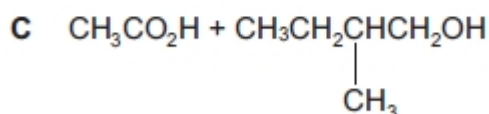
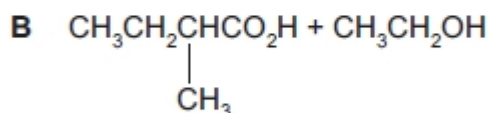
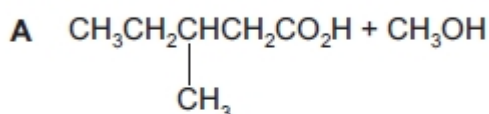
- 1 The name of this compound is ethyl butanoate.
- 2 This compound is a structural isomer of hexanoic acid.
- 3 When this compound is heated with aqueous sodium hydroxide, the products are butan-1-ol and sodium ethanoate.

Q# 1195/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 22//www.SmashingScience.org :o)

22 An ester with an odour of banana has the following formula.

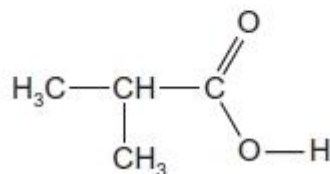
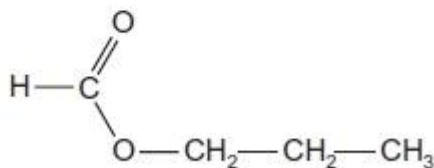
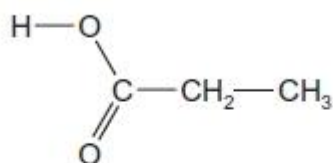
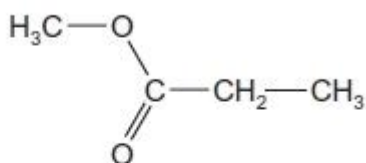


Which pair of reactants, under suitable conditions, will produce this ester?



Q# 1196/ AS Chemistry/2013/w/TZ 1/Paper 1/Q# 29//www.SmashingScience.org :o)

29 How many of the compounds shown will react with aqueous sodium hydroxide to form the sodium salt of a carboxylic acid?



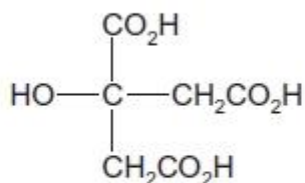
A 1

B 2

C 3

D 4

- 26 How many moles of hydrogen, H_2 , are evolved when an excess of sodium metal is added to one mole of citric acid?



citric acid

- A 1 B 2 C 3 D 4

- 6 An experiment is set up to measure the rate of hydrolysis of ethyl ethanoate.



The hydrolysis is found to be slow in neutral aqueous solution but it proceeds at a measurable rate when the solution is acidified with hydrochloric acid.

What is the function of the hydrochloric acid?

- A to dissolve the ethyl ethanoate
B to ensure that the reaction reaches equilibrium
C to increase the reaction rate by catalytic action
D to suppress ionisation of the ethanoic acid formed

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 40 Which reagents, when used in an excess, can be used to make sodium lactate, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{Na}$, from lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$?

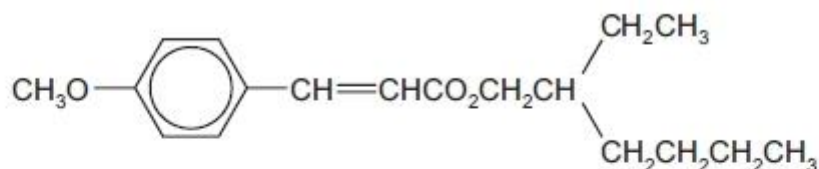
- 1 Na
2 NaHCO_3
3 NaOH

The responses **A** to **D** should be selected on the basis of

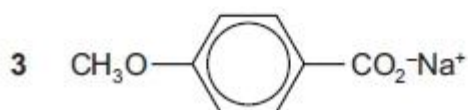
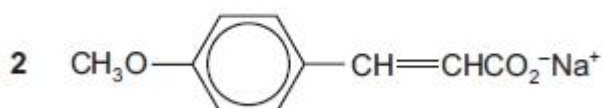
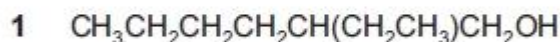
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



39 A sun protection cream contains the following ester as its active ingredient.

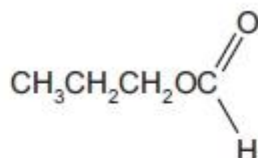


Which substances are present in the products of its hydrolysis by aqueous sodium hydroxide?



Q# 1201/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 The structural formula of a compound X is shown below.



What is the name of compound X and how does its boiling point compare with that of butanoic acid?

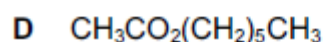
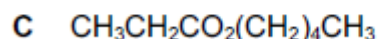
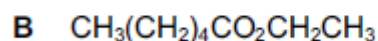
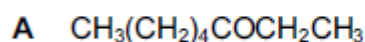
	name of X	boiling point of X
A	methyl propanoate	higher
B	methyl propanoate	lower
C	propyl methanoate	higher
D	propyl methanoate	lower

Q# 1202/ AS Chemistry/2012/w/TZ 1/Paper 1/Q# 30//www.SmashingScience.org :o)

30 Compound X, $\text{C}_6\text{H}_{12}\text{O}$, is oxidised by acidified sodium dichromate(VI) to compound Y.

Compound Y reacts with ethanol in the presence of a little concentrated sulfuric acid to give liquid Z.

What is the formula of Z?



Q# 1203/ AS Chemistry/2012/w/TZ 1/Paper 1/Q# 27//www.SmashingScience.org :o)

27 Which reagent reacts with ethanol and also reacts with ethanoic acid?

- A acidified potassium dichromate(VI)
- B sodium
- C sodium carbonate
- D sodium hydroxide

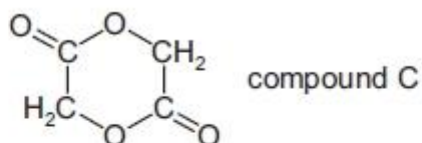
Q# 1204/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 How many isomeric esters have the molecular formula $C_4H_8O_2$?

- A 2
- B 3
- C 4
- D 5

Q# 1205/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

23 Compound C is used in textile and leather processing.



Which starting material(s), on gentle heating with a few drops of concentrated sulfuric acid, generates compound C?

- A CH_3COOH only
- B $HOCH_2COOH$ only
- C CH_3COOCH_2COOH only
- D CH_3COOH mixed with $HOCH_2COOH$

Q# 1206/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 22//www.SmashingScience.org :o)

22 Use of the Data Booklet is relevant to this question.

A sample of ethyl propanoate is hydrolysed by heating under reflux with aqueous sodium hydroxide. The two organic products of the hydrolysis are separated, purified and weighed.

Out of the total mass of products obtained, what is the percentage by mass of each product?

- A 32.4 % and 67.6 %
- B 38.3 % and 61.7 %
- C 42.3 % and 57.7 %
- D 50.0 % and 50.0 %

Q# 1207/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 21//www.SmashingScience.org :o)

21 An organic compound J reacts with sodium to produce an organic ion with a charge of -3 . J reacts with $NaOH(aq)$ to produce an organic ion with a charge of -1 .

What could be the structural formula of J?

- A $HO_2CCH(OH)CH_2CO_2H$
- B $HO_2CCH(OH)CH_2CHO$
- C $HOCH_2CH(OH)CH_2CO_2H$
- D $HOCH_2COCH_2CHO$

30 Which pair of substances could react to give the ester $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$?

- A ethanol and ethanoic acid
- B methanol and ethanoic acid
- C methanol and propanoic acid
- D propan-1-ol and methanoic acid

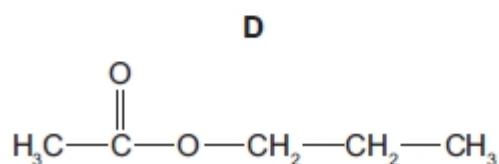
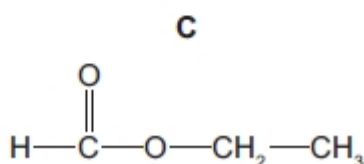
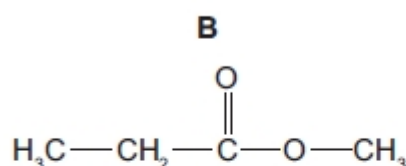
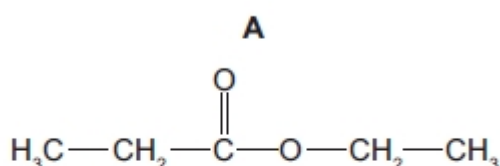
Q# 1209/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 22//www.SmashingScience.org :o)

22 In the general formula of which class of compound, is the ratio of hydrogen atoms to carbon atoms the highest?

- A alcohols
- B aldehydes
- C carboxylic acids
- D halogenoalkanes

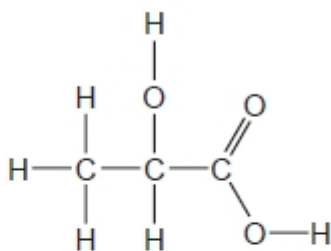
Q# 1210/ AS Chemistry/2011/s/TZ 1/Paper 1/Q# 27//www.SmashingScience.org :o)

27 Which formula represents an ester which will form sodium ethanoate on hydrolysis with aqueous sodium hydroxide?



Q# 1211/ AS Chemistry/2010/w/TZ 1/Paper 1/Q# 26//www.SmashingScience.org :o)

26 Lactic acid occurs naturally, for example in sour milk.



lactic acid

What is a property of lactic acid?

- A It decolourises aqueous bromine rapidly.
- B It is insoluble in water.
- C It reduces Fehling's reagent.
- D Two molecules react with each other in the presence of a strong acid.



22 A compound Y has all of the properties below.

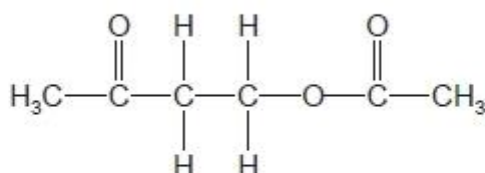
- It is a liquid at 25 °C.
- It mixes completely with water.
- It reacts with aqueous sodium hydroxide.

What could Y be?

- A ethanoic acid
B ethanol
C ethene
D ethyl ethanoate

Q# 1213/ AS Chemistry/2010/w/TZ 1/Paper 1/Q# 21//www.SmashingScience.org :o)

21 Compound X reacts with ethanoic acid in the presence of an H^+ catalyst to produce the compound below.



What is the molecular formula of compound X?

- A $C_2H_6O_2$ B $C_2H_6O_3$ C C_4H_8O D $C_4H_8O_2$

Q# 1214/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 34//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

34 When organic refuse decomposes in water carboxylic acids are formed. The water becomes acidic and aquatic life is destroyed.

Which additives are suitable to remove this acid pollution?

- 1 calcium carbonate
2 calcium hydroxide
3 potassium nitrate

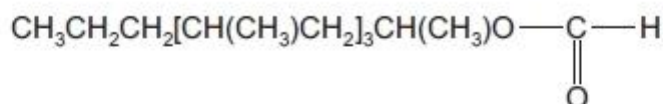
Q# 1215/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 Which compound is a product of the hydrolysis of $CH_3CO_2C_3H_7$ by boiling aqueous sodium hydroxide?

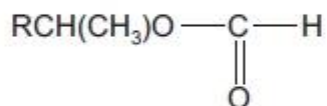
- A CH_3OH B C_3H_7OH C $C_3H_7CO_2H$ D $C_3H_7CO_2^- Na^+$



- 30 The acarid mite releases *lardolure* to attract other mites to a host. This chemical can be destroyed by hydrolysis with acid.



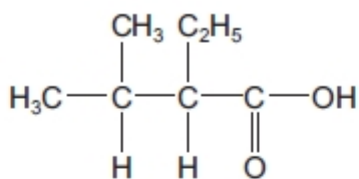
A simplified formula for *lardolure* may be written as follows.



What are the products of its hydrolysis?

- A $\text{RCH}(\text{CH}_3)\text{CO}_2\text{H} + \text{CH}_3\text{OH}$
- B $\text{RCH}(\text{CH}_3)\text{CO}_2\text{H} + \text{HCO}_2\text{H}$
- C $\text{RCH}(\text{CH}_3)\text{OH} + \text{CO}_2$
- D $\text{RCH}(\text{CH}_3)\text{OH} + \text{HCO}_2\text{H}$

- 29 The characteristic odour of rum is attributed to the compound 2-ethyl-3-methylbutanoic acid.



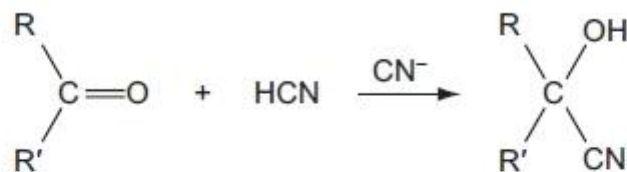
2-ethyl-3-methylbutanoic acid

Which compound will produce 2-ethyl-3-methylbutanoic acid by heating under reflux with alcoholic sodium cyanide and subsequent acid hydrolysis of the reaction product?

- A $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CHBr}-\text{CH}_3 \end{array}$
- B $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CHBr}-\text{CH}_2-\text{CH}_3 \end{array}$
- C $\begin{array}{c} \text{CH}_3 \quad \text{C}_2\text{H}_5 \\ | \quad | \\ \text{CH}_3-\text{CH}-\text{CH}-\text{CH}_2\text{Br} \end{array}$
- D $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2\text{Br} \\ | \quad | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \end{array}$



28 Cyanohydrins can be made from carbonyl compounds by generating CN^- ions from HCN in the presence of a weak base.



In a similar reaction, $^-\text{CH}_2\text{CO}_2\text{CH}_3$ ions are generated from $\text{CH}_3\text{CO}_2\text{CH}_3$ by strong bases.

Which compound can be made from an aldehyde and $\text{CH}_3\text{CO}_2\text{CH}_3$ in the presence of a strong base?

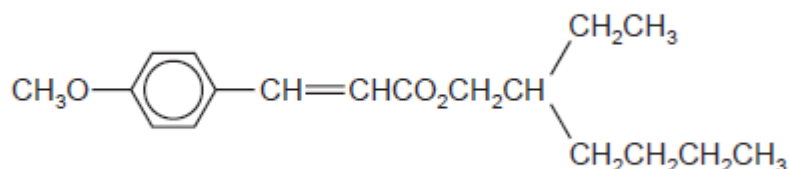
- A** $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{CH}_3$
- B** $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- C** $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CO}_2\text{CH}_3$
- D** $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CO}_2\text{CH}_3$

Q# 1219/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

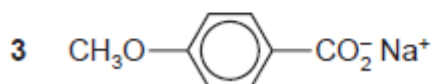
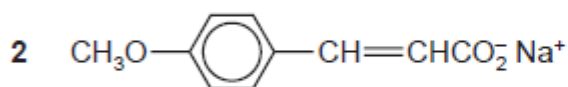
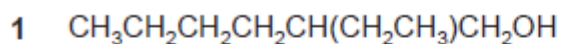
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 A sun protection cream contains the following ester as its active ingredient.



What are the products of its partial or total hydrolysis by aqueous sodium hydroxide?



30 A compound Y has the following properties.

- It is a liquid at room temperature and atmospheric pressure.
- It does not mix completely with water.
- It does not decolourise acidified potassium manganate(VII).

What could Y be?

- A ethane
- B ethanoic acid
- C ethanol
- D ethyl ethanoate

Q# 1221/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 27//www.SmashingScience.org :o)

27 Which ester is formed when the alcohol $\text{CH}_3\text{CH}_2\text{OH}$ is reacted with $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$?

- A ethyl propanoate
- B ethyl butanoate
- C propyl ethanoate
- D butyl ethanoate

Mark Scheme ALyl Chem 18 EQ P1 22w to 09s Paper 1 Carboxylic acids etc 94marks

Q# 1128/ AS Chemistry/2022/w/TZ 1/Paper 1/Q#

36//www.SmashingScience.org :o)

36 | D

Q# 1129/ AS Chemistry/2022/w/TZ 1/Paper 1/Q#

32//www.SmashingScience.org :o)

32 | A

Q# 1130/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

38//www.SmashingScience.org :o)

37 | B

Q# 1131/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

37//www.SmashingScience.org :o)

37 | B

Q# 1132/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

28//www.SmashingScience.org :o)

26 | C

Q# 1133/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

27//www.SmashingScience.org :o)

26 | C

Q# 1134/ AS Chemistry/2022/m/TZ 2/Paper 1/Q#

35//www.SmashingScience.org :o)

35 | A

Q# 1135/ AS Chemistry/2022/m/TZ 2/Paper 1/Q#

34//www.SmashingScience.org :o)

34 | C

Q# 1136/ AS Chemistry/2022/m/TZ 2/Paper 1/Q#

32//www.SmashingScience.org :o)

32 | A

Q# 1137/ AS Chemistry/2021/w/TZ 1/Paper 1/Q#

29//www.SmashingScience.org :o)

29 | B

Q# 1138/ AS Chemistry/2021/w/TZ 1/Paper 1/Q#

28//www.SmashingScience.org :o)

28 | C

Q# 1139/ AS Chemistry/2021/w/TZ 1/Paper 1/Q#

24//www.SmashingScience.org :o)

24 | C

Q# 1140/ AS Chemistry/2021/s/TZ 1/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | C

Q# 1141/ AS Chemistry/2021/s/TZ 1/Paper 1/Q#

24//www.SmashingScience.org :o)

24 | D

Q# 1142/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#

40//www.SmashingScience.org :o)

40 | B

Q# 1143/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#

29//www.SmashingScience.org :o)

29 | C



Q# 1144/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#
23//www.SmashingScience.org :o)

23 | A

Q# 1145/ AS Chemistry/2020/w/TZ 1/Paper 1/Q#
40//www.SmashingScience.org :o)

40 | B

Q# 1146/ AS Chemistry/2020/w/TZ 1/Paper 1/Q#
29//www.SmashingScience.org :o)

29 | D

Q# 1147/ AS Chemistry/2020/s/TZ 1/Paper 1/Q#
30//www.SmashingScience.org :o)

30 | A

Q# 1148/ AS Chemistry/2020/s/TZ 1/Paper 1/Q#
26//www.SmashingScience.org :o)

26 | D

Q# 1149/ AS Chemistry/2020/m/TZ 2/Paper 1/Q#
22//www.SmashingScience.org :o)

22 | B

Q# 1150/ AS Chemistry/2019/w/TZ 1/Paper 1/Q#
40//www.SmashingScience.org :o)

40 | D

Q# 1151/ AS Chemistry/2019/w/TZ 1/Paper 1/Q#
29//www.SmashingScience.org :o)

29 | C

Q# 1152/ AS Chemistry/2019/w/TZ 1/Paper 1/Q#
28//www.SmashingScience.org :o)

28 | B

Q# 1153/ AS Chemistry/2019/s/TZ 1/Paper 1/Q#
40//www.SmashingScience.org :o)

40 | C

Q# 1154/ AS Chemistry/2019/s/TZ 1/Paper 1/Q#
39//www.SmashingScience.org :o)

39 | C

Q# 1155/ AS Chemistry/2019/m/TZ 2/Paper 1/Q#
40//www.SmashingScience.org :o)

40 | C

Q# 1156/ AS Chemistry/2019/m/TZ 2/Paper 1/Q#
29//www.SmashingScience.org :o)

29 | D

Q# 1157/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#
29//www.SmashingScience.org :o)

29 | C

Q# 1158/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#
28//www.SmashingScience.org :o)

28 | B

Q# 1159/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#
26//www.SmashingScience.org :o)

26 | D

Q# 1160/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#
24//www.SmashingScience.org :o)

24 | D

Q# 1161/ AS Chemistry/2018/s/TZ 1/Paper 1/Q#
29//www.SmashingScience.org :o)

29 | C

Q# 1162/ AS Chemistry/2018/s/TZ 1/Paper 1/Q#
28//www.SmashingScience.org :o)

28 | A

Q# 1163/ AS Chemistry/2018/s/TZ 1/Paper 1/Q#
27//www.SmashingScience.org :o)

27 | A

Q# 1164/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#
40//www.SmashingScience.org :o)

40 | B

Q# 1165/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#
29//www.SmashingScience.org :o)

29 | C

Q# 1166/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#
28//www.SmashingScience.org :o)

28 | D

Q# 1167/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 30//

30 | C

Q# 1168/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 28//

28 | B

Q# 1169/ AS Chemistry/2017/s/TZ 1/Paper 1/Q#
40//www.SmashingScience.org :o)

40 | A

Q# 1170/ AS Chemistry/2017/s/TZ 1/Paper 1/Q#
24//www.SmashingScience.org :o)

24 | C

Q# 1171/ AS Chemistry/2017/m/TZ 2/Paper 1/Q#
40//www.SmashingScience.org :o)

40 | A

Q# 1172/ AS Chemistry/2017/m/TZ 2/Paper 1/Q#
30//www.SmashingScience.org :o)

30 | C

Q# 1173/ AS Chemistry/2016/w/TZ 1/Paper 1/Q#
39//www.SmashingScience.org :o)

39 | D

Q# 1174/ AS Chemistry/2016/w/TZ 1/Paper 1/Q#
30//www.SmashingScience.org :o)

30 | B

Q# 1175/ AS Chemistry/2016/w/TZ 1/Paper 1/Q# 29//:o)

29 | A



Q# 1176/ AS Chemistry/2016/s/TZ 1/Paper 1/Q# 29//:o)
 29 **A**
Q# 1177/ AS Chemistry/2016/s/TZ 1/Paper 1/Q# 27//:o)
 27 **B**
Q# 1178/ AS Chemistry/2016/s/TZ 1/Paper 1/Q# 26//:o)
 26 **C**
Q# 1179/ AS Chemistry/2016/s/TZ 1/Paper 1/Q# 21//:o)
 21 **D**
Q# 1180/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 39//:o)
 39 **A**
Q# 1181/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 37//:o)
 37 **C**
Q# 1182/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 30//:o)
 30 **B**
Q# 1183/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 28//:o)
 28 **C**
Q# 1184/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 27//:o)
 27 **D**
Q# 1185/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 39//:o)
 39 **D**
Q# 1186/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 37//:o)
 37 **C**
Q# 1187/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 30//:o)
 30 **B**
Q# 1188/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 20//:o)
 20 **C**
Q# 1189/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 40//:o)
 40 **C**
Q# 1190/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 38//:o)
 38 **C**
Q# 1191/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 28//:o)
 28 **C**
Q# 1192/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 27//:o)
 27 **A**
Q# 1193/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 40//:o)
 40 **A**
Q# 1194/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 38//:o)
 38 **B**
Q# 1195/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 22//:o)
 22 **C**
Q# 1196/ AS Chemistry/2013/w/TZ 1/Paper 1/Q# 29//:o)
 29 **D**
Q# 1197/ AS Chemistry/2013/w/TZ 1/Paper 1/Q# 26//:o)
 26 **B**
Q# 1198/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 6//:o)
 6 **C**

Q# 1199/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 40//:o)
 40 **C**
Q# 1200/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 39//:o)
 39 **B**
Q# 1201/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 28//:o)
 28 **D**
Q# 1202/ AS Chemistry/2012/w/TZ 1/Paper 1/Q# 30//:o)
 30 **B**
Q# 1203/ AS Chemistry/2012/w/TZ 1/Paper 1/Q# 27//:o)
 27 **B**
Q# 1204/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 24//:o)
 24 **C**
Q# 1205/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 23//:o)
 23 **B**
Q# 1206/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 22//:o)
 22 **A**
Q# 1207/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 21//:o)
 21 **C**
Q# 1208/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 30//:o)
 30 **C**
Q# 1209/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 22//:o)
 22 **A**
Q# 1210/ AS Chemistry/2011/s/TZ 1/Paper 1/Q# 27//:o)
 27 **D**
Q# 1211/ AS Chemistry/2010/w/TZ 1/Paper 1/Q# 26//:o)
 26 **D**
Q# 1212/ AS Chemistry/2010/w/TZ 1/Paper 1/Q# 22//:o)
 22 **A**
Q# 1213/ AS Chemistry/2010/w/TZ 1/Paper 1/Q# 21//:o)
 21 **D**
Q# 1214/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 34//:o)
 34 **B**
Q# 1215/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 25//:o)
 25 **B**
Q# 1216/ AS Chemistry/2009/w/TZ 1/Paper 1/Q# 30//:o)
 30 **D**
Q# 1217/ AS Chemistry/2009/w/TZ 1/Paper 1/Q# 29//:o)
 29 **B**
Q# 1218/ AS Chemistry/2009/w/TZ 1/Paper 1/Q# 28//:o)
 28 **C**
Q# 1219/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 40//:o)
 40 **B**
Q# 1220/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 30//:o)
 30 **D**
Q# 1221/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 27//:o)
 27 **B**

